Davide Bortoletto

GIS Analyst / GIS Specialist

Padua, Italy
d.bortoletto17@gmail.com
Portfolio: dav9817.github.io · LinkedIn: linkedin.com/in/davide-bortoletto-889a36301

PROFESSIONAL PROFILE

GIS Analyst with a background in urban and regional planning (Master's Degree, LM-48, IUAV University of Venice) and professional experience in geospatial analysis, cartographic production and the integration of georeferenced data. Skilled in multi-criteria decision analysis (MCDA), remote sensing and satellite data, LiDAR processing, process automation in Python and the publication of results as interactive web maps. Able to turn heterogeneous territorial data into maps and analyses that support decision-making.

SELECTED GIS PROJECTS

Utility-scale solar PV siting — Province of Verona
Multi-criteria decision analysis (MCDA) to delineate the macro-areas most suitable for utility-scale photovoltaic development. Land-use suitability, DEM-derived morphology (slope and aspect) and boolean exclusion masks (Natura 2000 network, water bodies, dense urban areas) are combined over a hexagonal grid of 14,756 cells. The output is a Solar Hotspot map condensed into three strategic districts, supporting investment decisions.

Vegetation risk on a high-voltage power line — Vallelaghi (Trento)
Simulation of a new overhead HV line and assessment of its interference with the surrounding forest. From LiDAR data, a geometric model of tree-fall risk on the conductors is built; three management strategies are compared by measuring residual risk over five years, showing the superiority of the predictive geometric approach.

NO2 and night-time lights in the Po Basin
Study of the relationship between nitrogen dioxide pollution and urbanisation in one of Europe's most polluted areas. A reproducible pipeline was built on Sentinel-5P/TROPOMI and Suomi-NPP/VIIRS data: cloud extraction on Google Earth Engine, statistical analysis in Python, cartography in QGIS. Night-time lights explain about 41% of the spatial variability of NO2; the remainder is driven by the shape of the Po Valley basin.

Settlement suitability on Mars — MCDA on NASA/USGS data
Multi-criteria analysis in QGIS to identify the areas most suitable for a future human base on Mars, from an engineering rather than a biological perspective. Five criteria reclassified on a common 0–5 scale: terrain morphology, access to subsurface water ice, surface geology, insolation and mean temperature. The model yields a composite suitability map, a shortlist of candidate sites and a functional zoning, published as an interactive 3D globe.

TECHNICAL SKILLS

- **GIS & Geospatial Analysis:** QGIS, multi-criteria analysis (MCDA, weighted overlay), geoprocessing, buffer analysis, zonal statistics, raster reclassification, geodatabase management, cartographic production; integration of georeferenced data (Veneto Geoportal, municipal SIT, NASA/USGS open data); basic knowledge of ArcGIS (academic use).
- **Remote Sensing:** Google Earth Engine; Copernicus/Sentinel-5P (TROPOMI) and Suomi-NPP/VIIRS data; eCognition (Object-Based Image Analysis - OBIA), ERMapper.
- **LiDAR & Terrain Models:** point-cloud processing, DEM/DSM, morphological analysis (slope, aspect).
- **Programming & Automation:** Python (scripting for GIS process automation and statistical analysis), GDAL/OGR.
- **Spatial Databases:** PostgreSQL/PostGIS and spatial SQL (in training).
- **Web Mapping & Publishing:** interactive web maps, vector tiles, HTML/CSS/JavaScript.
- **CAD & Technical Drafting:** Autodesk AutoCAD, ArchiCAD, 2D technical drawing.
- **Graphics & Image Editing:** Adobe Photoshop, Adobe Lightroom, GIMP, Pixelmator Pro.
- **Productivity & Systems:** Microsoft Office (Word, Excel, PowerPoint); Windows, macOS.

PROFESSIONAL EXPERIENCE

GIS Analyst & CAD Draftsman | TERRE S.r.l.
Mestre (Venice), Italy · 02/2025 – 03/2026 (freelance / VAT collaboration)

- Developed geospatial analyses in QGIS to define buffer zones around Palladian villas, supporting landscape-protection studies.

- Collected, managed and integrated georeferenced data from the Veneto Geoportal, producing new thematic datasets.
- Developed Python scripts to automate and streamline GIS processes.
- Carried out preliminary cartographic, historical and urban-planning analyses to support territorial planning.
- Contributed to post-works environmental and landscape impact assessment sheets (photographic documentation and urban-planning drawings) for major infrastructure projects (e.g. SPV).
- Produced 2D CAD deliverables (green-works site plans for infrastructure) using external references.
- Edited and post-processed images and maps with graphics software (Photoshop, GIMP).

Intern | Studio Negri & Fauro Architetti Associati
Padua, Italy · 06/2022 – 07/2022

- Prepared site plans (scale 1:200 – 1:2000) and conducted field surveys, reviewing municipal planning documents, constraint maps and land-transformability frameworks.
- Produced CAD technical drawings of plans, sections and elevations (scale 1:100) for energy-efficiency renovation projects.
- Verified urban-planning compliance by reviewing building regulations and municipal plans (Valbrenta, Abano Terme, Noventa di Piave).
- Produced graphic restitutions of building surveys.

Intern | Municipality of San Giorgio delle Pertiche
Padua, Italy · 07/2019 – 08/2019

- Consulted cadastral data for taxation purposes through the municipal Territorial Information System (SIT/GIS).
- Checked the tax and contribution compliance of commercial businesses via institutional portals (INAIL, Telemaco).
- Built and maintained a structured registry (Excel database) and handled certified email (PEC).

Trainee | Studio Tecnico Associato Battiston e Facco
Padua, Italy · 06/2017 – 08/2017

- Assisted with surveying and CAD graphic restitution of buildings following on-site inspections.

EDUCATION

Master's Degree (Laurea Magistrale, LM-48) in City and Environment: Planning and Policies — Università IUAV di Venezia
Venice, Italy · 2020 – 2024 · Final grade: 106/110
Main field of study: regional, urban and environmental planning (ISCED 0731). Curriculum: City–Territory–Landscape. Coursework in GIScience, geographic information systems and remote sensing, territorial and landscape planning, and land governance.

Bachelor's Degree (Laurea Triennale, L-21) in Land and Landscape Restoration and Enhancement — University of Padua
Padua, Italy · 2017 – 2020 · Final grade: 103/110
Main field of study: town, regional and environmental planning (ISCED 0731). Curriculum: Land Protection and Management (TRT). Coursework in GIS, CAD, territorial representation, hydrogeological risk and land planning.

Technical High School Diploma — "G. Belzoni" Technical Institute, Padua
Padua, Italy · 2013 – 2017 · Final grade: 80/100

CERTIFICATIONS

- Remote pilot of unmanned aircraft systems (UAS) — "open" category, subcategories A1/A3 (EASA/ENAC, Reg. (EU) 2019/947). Certificate no. ITA-RP-000081170acx, valid until 13/07/2031.

LANGUAGES

- Italian — Native.
- English — B1 level (listening, reading, speaking, writing).
- Driving licence — Categories A and B.